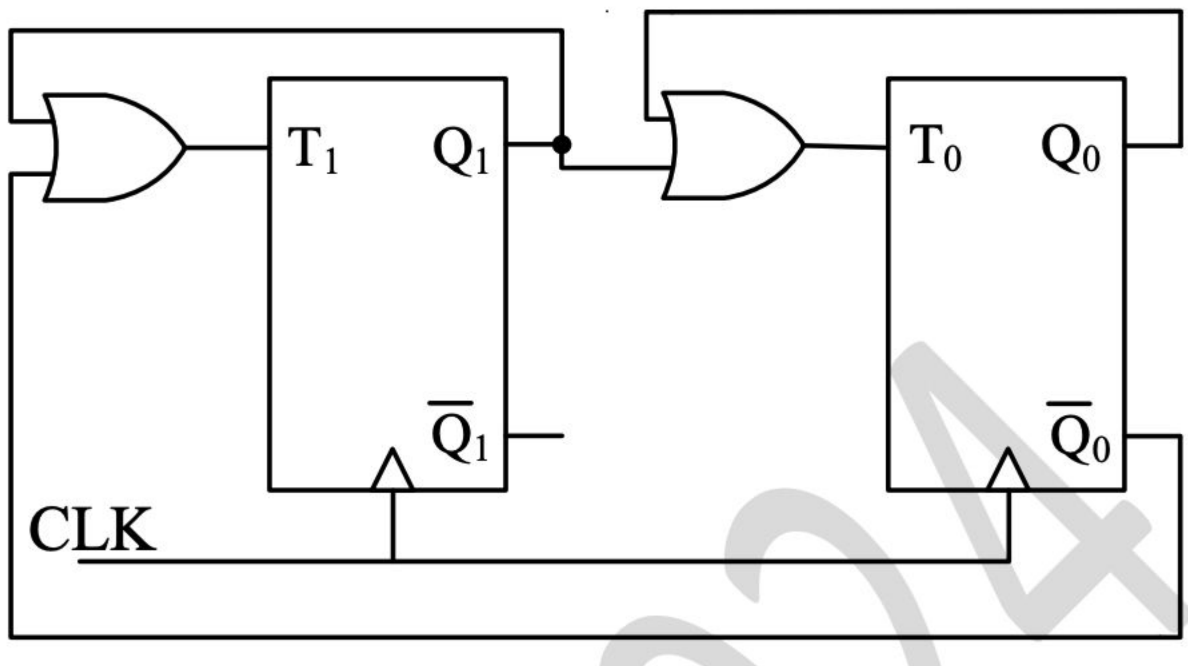
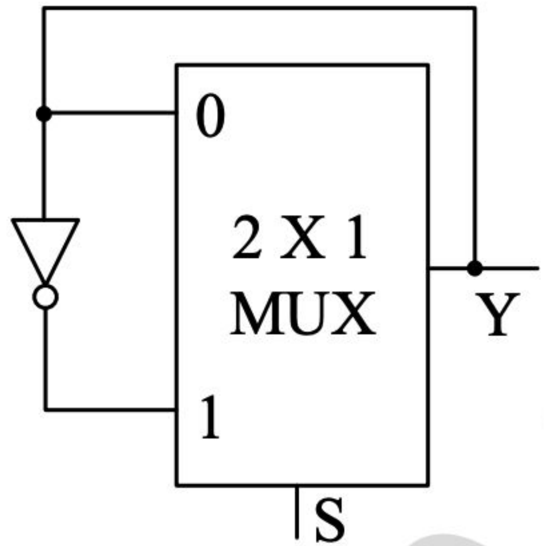
The sequence of states (Q_1Q_0) of the given synchronous sequential circuit is ________.

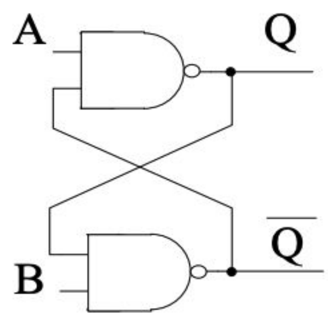


The propagation delay of the 2×1 MUX shown in the circuit is $10\,\text{ns}$. Consider the propagation delay of the inverter as $0\,\text{ns}$.

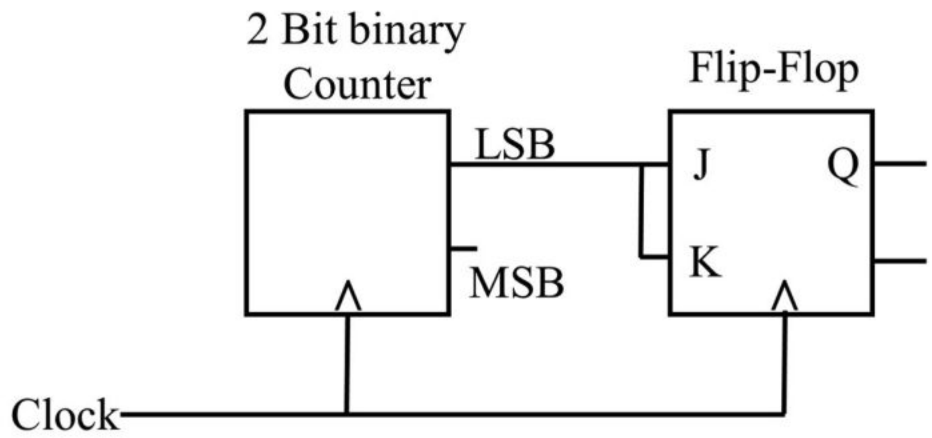


If S is set to 1 then the output Y is _______.

For the circuit shown below, the propagation delay of each NAND gate is 1 ns. The critical path delay, in ns, is __________ (*rounded off to the nearest integer*).

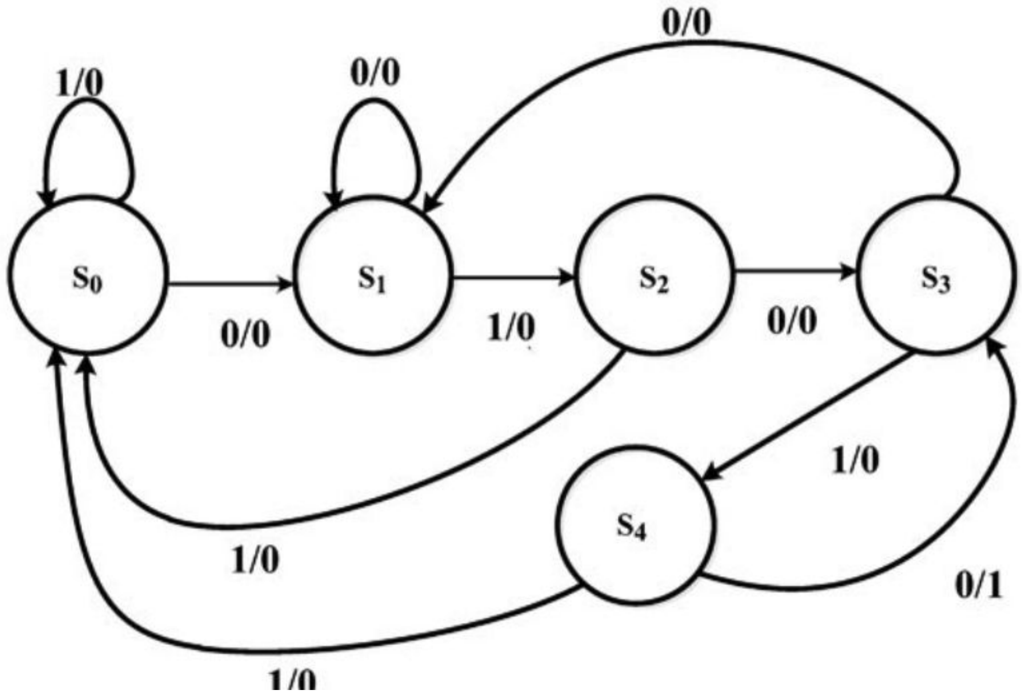


For the circuit shown, the clock frequency is f_0 and the duty cycle is 25%. For the signal at the Q output of the Flip-Flop, _______.

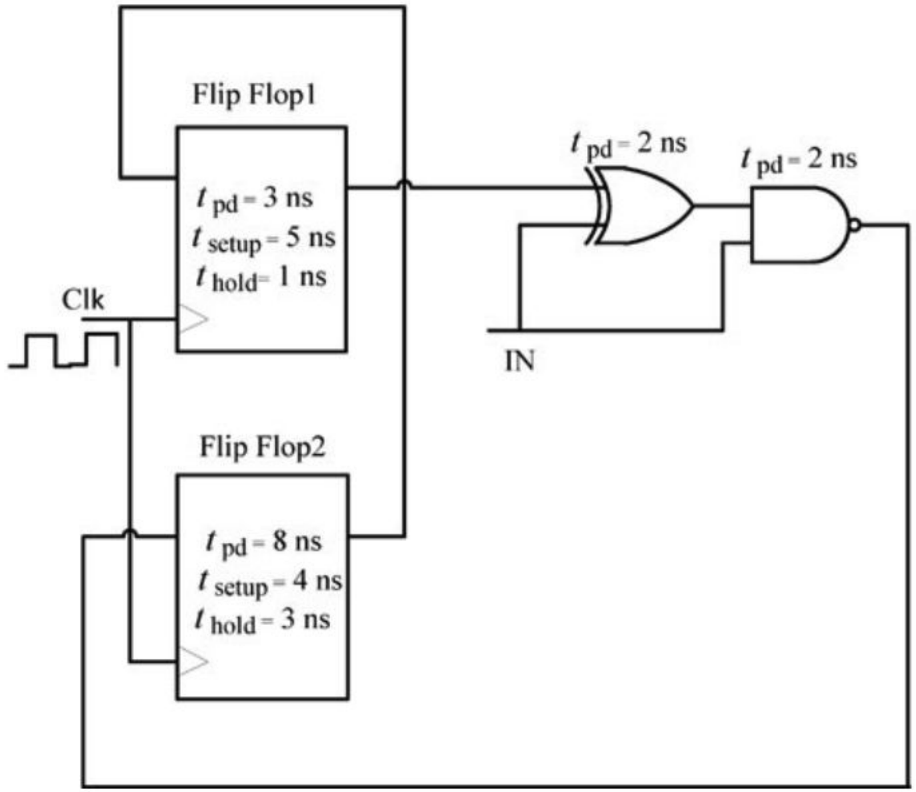


Consider a Boolean gate (D) where the output Y is related to the inputs A and B as, $Y = A + \bar{B}$, where $+$ denotes logical OR operation. The Boolean inputs '0' and '1' are also available separately. Using instances of only D gates and inputs '0' and '1', __________ (select the correct option(s)).

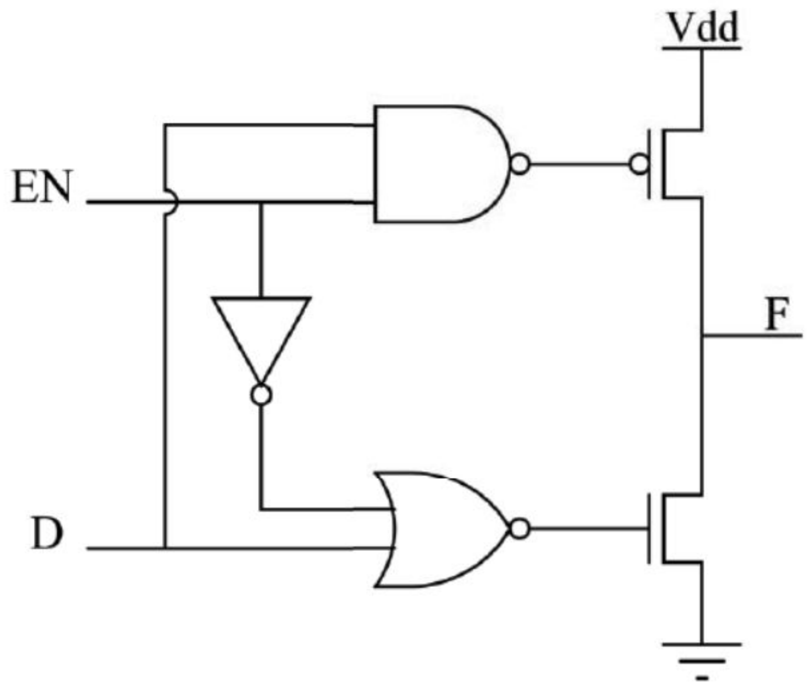
The state diagram of a sequence detector is shown below. State S_0 is the initial state of the sequence detector. If the output is 1, then



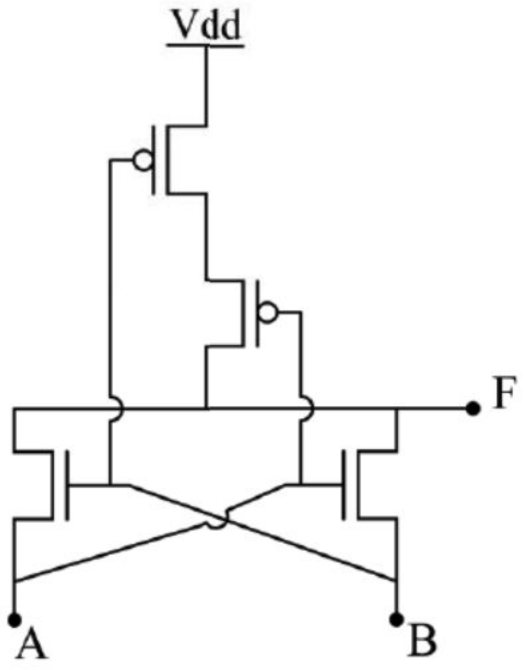
For the components in the sequential circuit shown below, t_{pd} is the propagation delay, t_{setup} is the setup time, and t_{hold} is the hold time. The maximum clock frequency **(rounded off to the nearest integer)**, at which the given circuit can operate reliably, is ________ MHz.



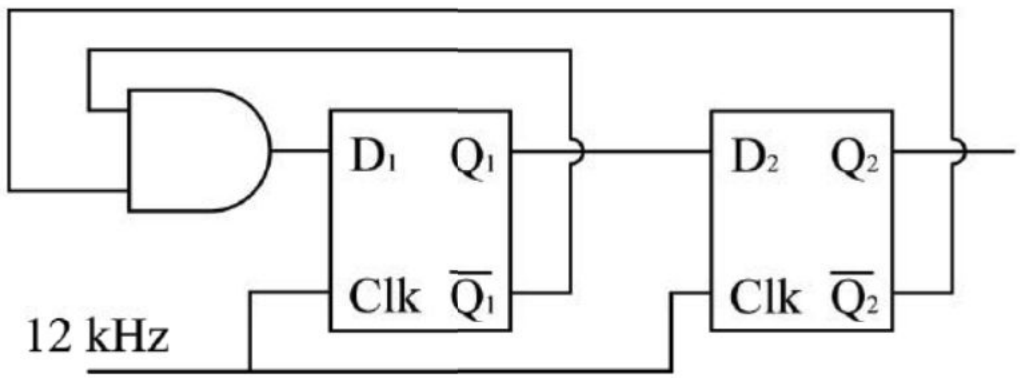
In the circuit shown, what are the values of F for EN = 0 and EN = 1, respectively?



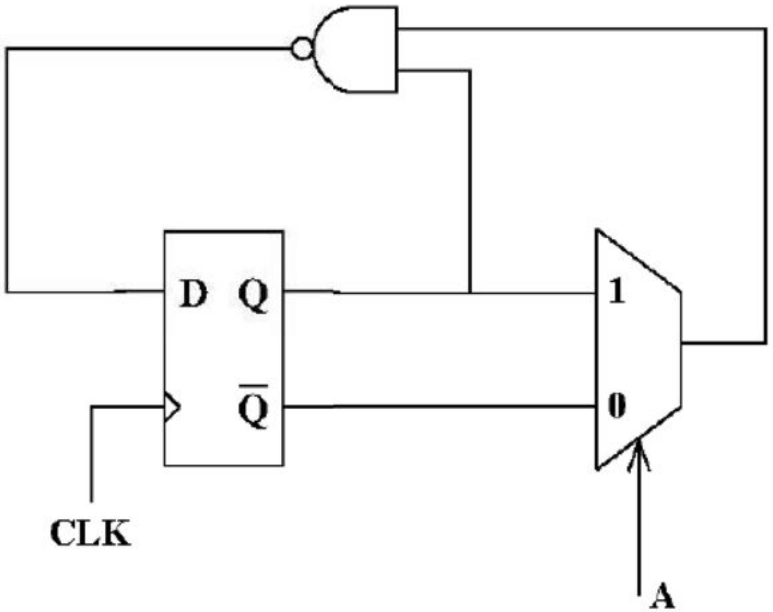
In the circuit shown, A and B are the inputs and F is the output. What is the functionality of the circuit?



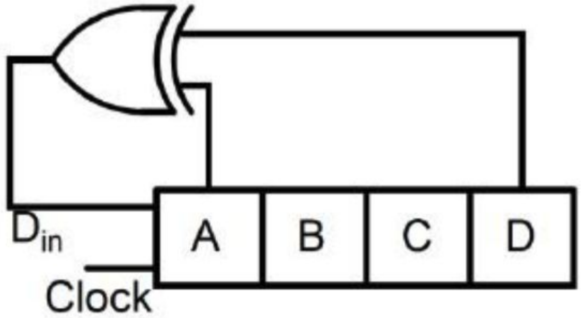
In the circuit shown, the clock frequency, i.e., the frequency of the Clk signal, is 12 kHz. The frequency of the signal at Q_2 is ____ kHz.



The state transition diagram for the circuit shown is



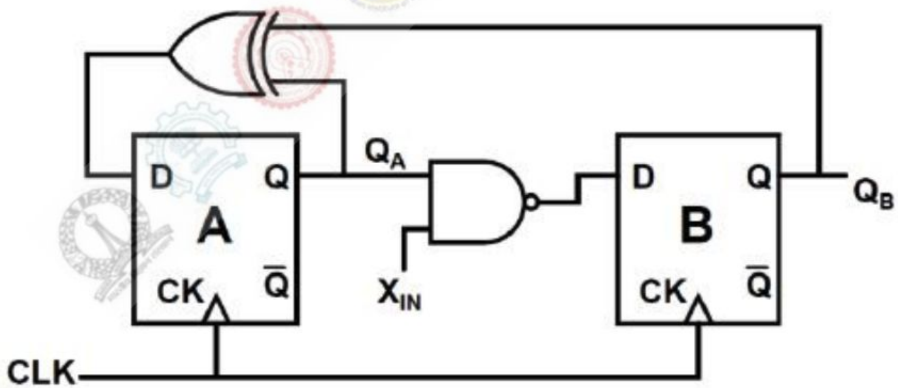
A 4-bit shift register circuit configured for right-shift operation, i.e. $D_{in} \rightarrow A,\ A \rightarrow B,\ B \rightarrow C,\ C \rightarrow D$, is shown. If the present state of the shift register is $ABCD = 1101$, the number of clock cycles required to reach the state $ABCD = 1111$ is ____________.

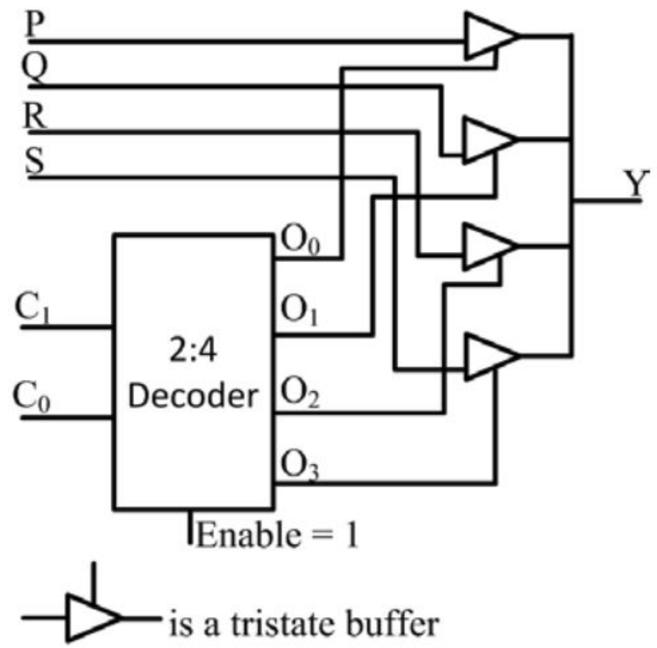


Which one of the following gives the simplified sum of products expression for the Boolean function $F = m_0 + m_2 + m_3 + m_5$, where m_0, m_2, m_3 and m_5 are minterms corresponding to the inputs A, B and C with A as the MSB and C as the LSB?

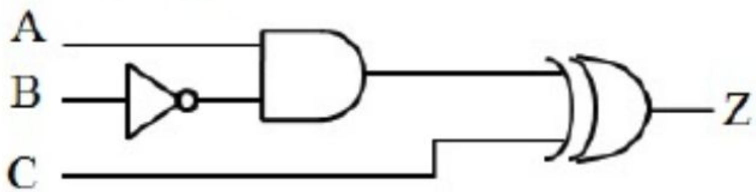
(A) $\bar{A}B + \bar{A}\bar{B}\bar{C} + A\bar{B}C$
(B) $\bar{A}\bar{C} + \bar{A}B + A\bar{B}C$
(C) $\bar{A}\bar{C} + A\bar{B} + A\bar{B}C$
(D) $\bar{A}BC + \bar{A}\bar{C} + A\bar{B}C$

A finite state machine (FSM) is implemented using the D flip-flops A and B, and logic gates, as shown in the figure below. The four possible states of the FSM are $Q_A Q_B = 00, 01, 10,$ and 11.

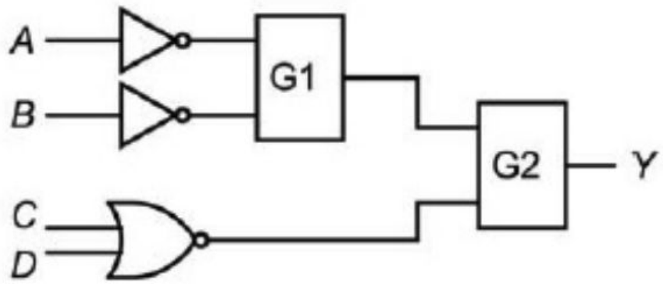



P
Q
R
S
Y
O_0
O_1
O_2
O_3
C_1
C_0
2:4 Decoder
Enable = 1
is a tristate buffer

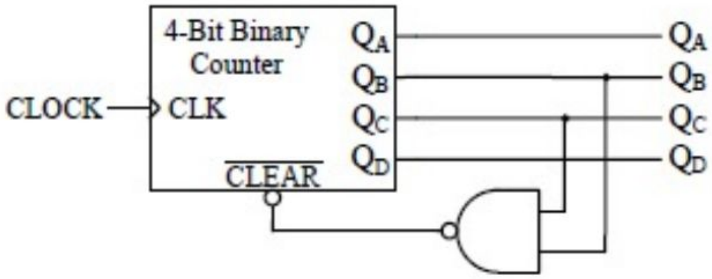
All the logic gates shown in the figure have a propagation delay of 20 ns. Let A = C = 0 and B = 1 until time t = 0. At t = 0, all the inputs flip (i.e., A = C = 1 and B = 0) and remain in that state. For t > 0, output Z = 1 for a duration (in ns) of .



In the figure shown, the output Y is required to be $Y = A\,B + \bar{C}\,\overline{D}$. The gates G1 and G2 must be, respectively,



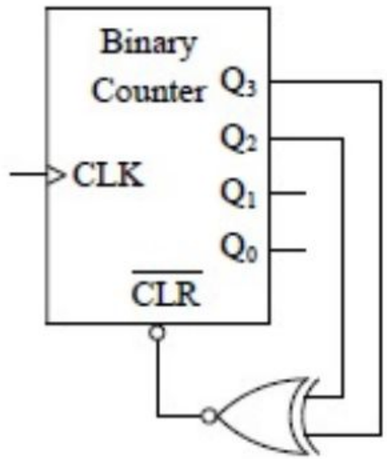
A mod-n counter using a synchronous binary up-counter with synchronous clear input is shown in the figure. The value of n is _______.



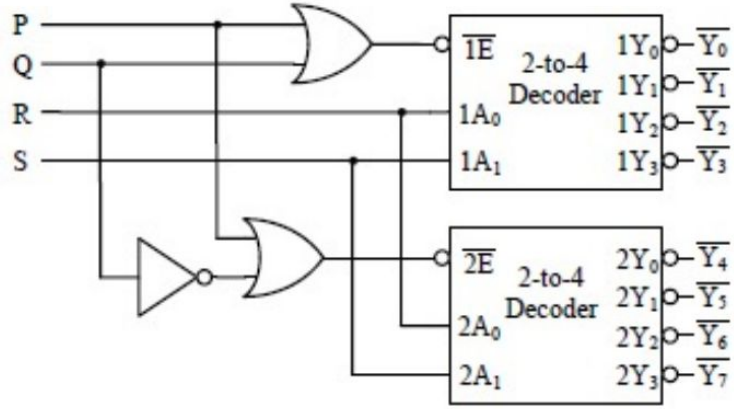
A function of Boolean variables X, Y and Z is expressed in terms of the min-terms as

$$F(X, Y, Z) = \Sigma\,(1, 2, 5, 6, 7)$$

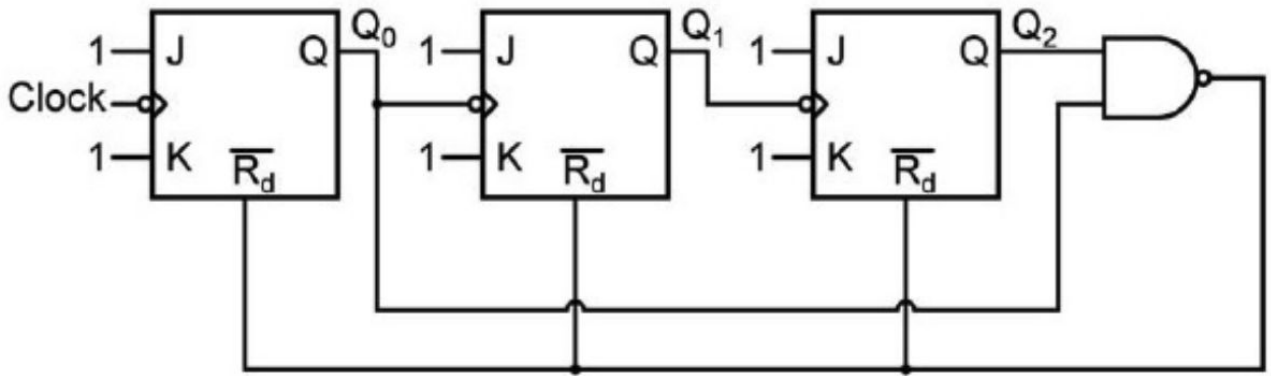
Which one of the product of sums given below is equal to the function $F(X, Y, Z)$?


Binary
Counter
Q_3
Q_2
Q_1
Q_0
CLK
$\overline{CLR}$

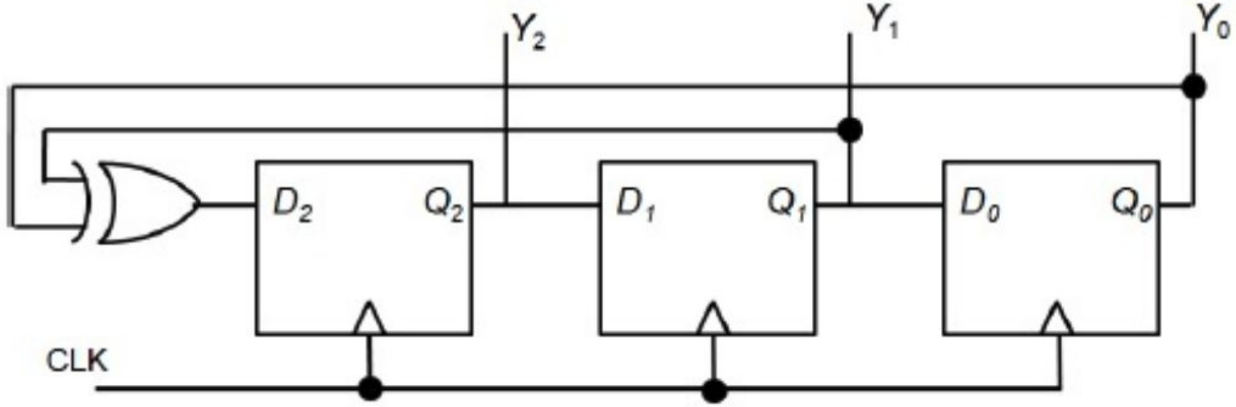
A 1-to-8 demultiplexer with data input D_{in} , address inputs S_0, S_1, S_2 (with S_0 as the LSB) and $\overline{Y}_0$ to $\overline{Y}_7$ as the eight demultiplexed outputs, is to be designed using two 2-to-4 decoders (with enable input $\bar{E}$ and address inputs A_0 and A_1) as shown in the figure. D_{in} , S_0, S_1 and S_2 are to be connected to P, Q, R and S, but not necessarily in this order. The respective input connections to P, Q, R, and S terminals should be



The circuit shown consists of J-K flip-flops, each with an active low asynchronous reset ($\overline{R_d}$ input). The counter corresponding to this circuit is



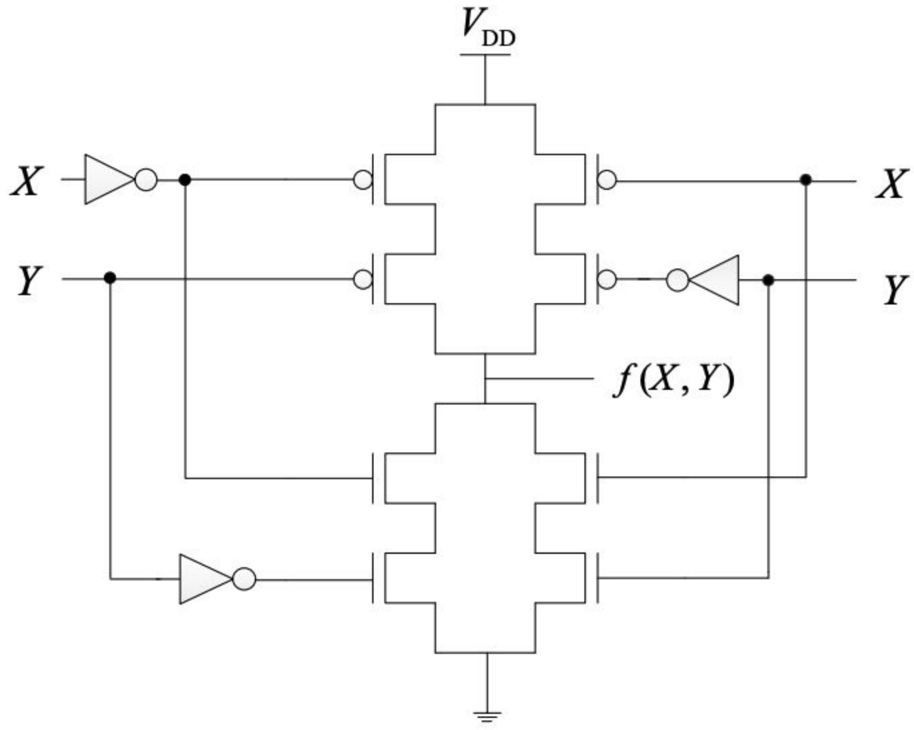
A three bit pseudo random number generator is shown. Initially the value of output $Y \equiv Y_2\, Y_1\, Y_0$ is set to 111. The value of output Y after three clock cycles is



A universal logic gate can implement any Boolean function by connecting sufficient number of them appropriately. Three gates are shown.

Gate 1 Gate 2 Gate 3

The logic function $f(X, Y)$ realized by the given circuit is



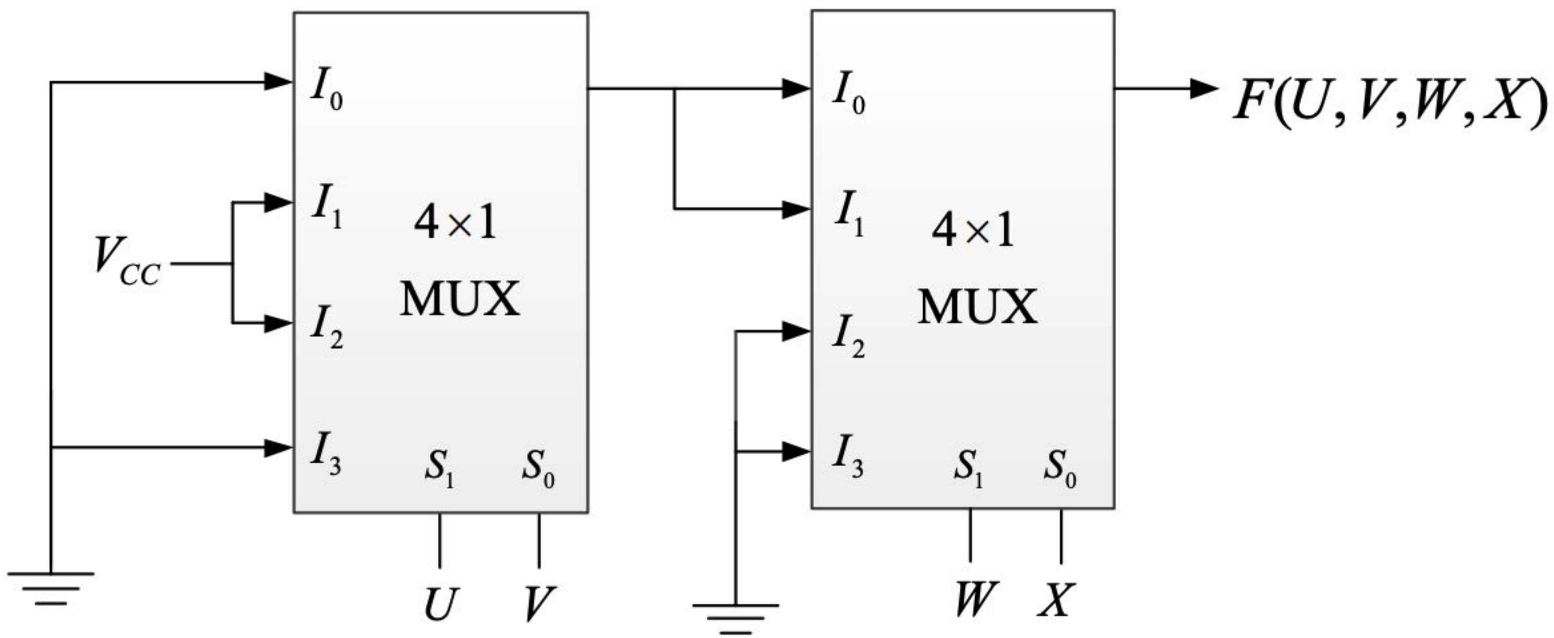
A function $F(A, B, C)$ defined by three Boolean variables A, B and C when expressed as sum of products is given by

$$F = \bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{A} \cdot B \cdot \bar{C} + A \cdot \bar{B} \cdot \bar{C}$$

where, $\bar{A}, \bar{B},$ and $\bar{C}$ are the complements of the respective variables. The product of sums (POS) form of the function F is

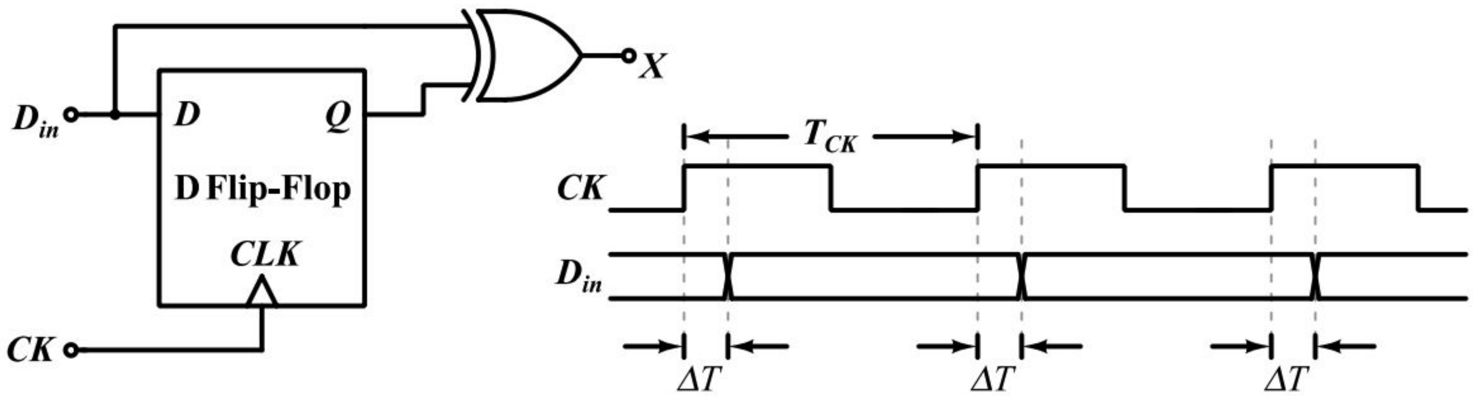
(A) $F = (A + B + C) \cdot (A + \bar{B} + C) \cdot (\bar{A} + B + C)$
(B) $F = (\bar{A} + \bar{B} + \bar{C}) \cdot (\bar{A} + B + \bar{C}) \cdot (A + \bar{B} + \bar{C})$
(C) $F = (A + B + \bar{C}) \cdot (A + \bar{B} + \bar{C}) \cdot (\bar{A} + B + \bar{C}) \cdot (\bar{A} + \bar{B} + C) \cdot (\bar{A} + \bar{B} + \bar{C})$
(D) $F = (\bar{A} + \bar{B} + C) \cdot (\bar{A} + B + C) \cdot (A + \bar{B} + C) \cdot (A + B + \bar{C}) \cdot (A + B + C)$

A four-variable Boolean function is realized using 4×1 multiplexers as shown in the figure.



The minimized expression for $F(U,V,W,X)$ is

In the circuit shown below, a positive edge-triggered D Flip-Flop is used for sampling input data D_{in} using clock CK. The XOR gate outputs 3.3 volts for logic HIGH and 0 volts for logic LOW levels. The data bit and clock periods are equal and the value of $\Delta T / T_{CK} = 0.15$, where the parameters ΔT and T_{CK} are shown in the figure. Assume that the Flip-Flop and the XOR gate are ideal.



If the probability of input data bit (D_{in}) transition in each clock period is 0.3, the average value (in volts, accurate to two decimal places) of the voltage at node X, is _______.